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A Scalable AI-Based System for Dynamic Traffic Signal Optimization and Real-Time Parking Space Detection

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Abstract—As cities grow rapidly and the number of cars on the road increases, a problem of traffic congestion and lack of parking spots has become one of the most critical issues in the contemporary cities. Conventional traffic control systems are usually based on the fixed or semi-adaptive control means and are not responsive to the real time situation. In this paper, a smart AI-based traffic and parking management system will be introduced, which combines machine learning and computer vision algorithms to overcome these challenges. The system is based on the Random Forest algorithm to process historical and live traffic data and predict congestion correctly and control traffic lights dynamically. Moreover, they use a computer vision module to track parking spots and see free parking lots in real-time, enabling users to find and book parking spots more effectively. The system proposed will enhance the general traffic, shorten the travel time, minimise congestion and give preference to emergency vehicles. Moreover, it offers a solution that is scalable and cost-effective and can be used in smart city transportation networks.

Index Terms—Predicting Traffic, Smart Parking, Machine Learning, Random Forest Algorithm, Intelligent Transportation Systems (ITS), Computer Vision, Real-Time Traffic Monitoring, Emergency Vehicle Prioritization, Smart Cities.

I. INTRODUCTION

The high rate of urbanization and the ever rising number of vehicles has contributed greatly to the congestion and car parking problems that have affected contemporary cities. Conventional traffic management systems are normally based on fixed signal timings or manual control, which cannot be adjusted to the real-time and unpredictable traffic conditions. Consequently, commuters have to face extended commuting time, more fuel usage, more pollution, and slower emergency services.

Over the past few years, new opportunities have been created by the creation of intelligent transportation systems due to the advances in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision. Traffic management solutions

can be made more adaptive, data-driven, and efficient with the help of these technologies. Machine learning algorithms can compute historical and real-time traffic data to forecast traffic congestion trends, traffic signal timing optimization, and route optimization. Meanwhile, computer vision can be used to monitor parking spaces automatically and enable users to find parking spots within seconds, minimizing the time they spend searching.

This paper suggests a smart traffic and parking control system, which will combine the traffic forecasting, based on the algorithm of the Random Forest, with the real-time parking space identification. The system gathers information through various sources including traffic cameras, IoT gadgets, and past traffic data to come up with precise congestion forecasts. It also offers intelligent parking suggestions, and enables users to book parking spaces beforehand. Moreover, adaptive signal control and route optimization are also applied to guarantee easier traffic flow and give precedence to emergency vehicles to respond to them faster.

The proposed solution will integrate both capabilities into one platform as opposed to many of the existing systems, which only undertake one or the other functionality (traffic prediction and parking management). This combined solution promotes real-time decision-making and user engagement, which helps to achieve better traffic flow, less overcrowding, better use of the parking facilities, and the creation of sustainable and intelligent urban transport systems.

II. LITERATURE REVIEW

The recent studies have discussed some intelligent strategies to solve the issue of traffic congestion with the help of sophisticated algorithms and intelligent technologies. A traffic management system that incorporates reinforcement learning and non-dominated sorting genetic algorithm was presented by Al-Heety et al. (2026) [1]. They apply adaptive Q-learning to

dynamically manage the traffic lights, prioritizing emergency vehicles. The system proved to be very effective in terms of traffic flow and lessening delays in urban settings with multiple intersections.

The next-generation smart traffic management framework is proposed by Goswami and Kadao (2025) [2] and is designed to be implemented in metropolitan cities of India. Their system uses a combination of IoT sensors, artificial intelligence, and machine learning to observe the conditions on the traffic and dynamically change the signal timings. Predictive analytics that were used were able to reduce congestion, travel time and vehicle emissions.

Xu and Li (2025) [3] created a real-time traffic jams prediction model that utilizes a multi-source fusion of data and dynamic risk field theory. Their system combined deep learning methods with intelligible models; hence, successfully capturing intricate traffic patterns and proactively controlling traffic.

Prathiba et al. (2025) [4] proposed a digital twin-based traffic and power grid management system in smart cities. Their methodology makes use of real-time data processing and predictive analytics to model urban infrastructure behavior. The model enhances efficiency of resources and increases overall traffic efficacy.

Wairagade et al. (2025) [5] suggested an AI-based traffic management system, which utilizes machine learning and computer vision to track and manage traffic in a city. Their system will gather real-time sensor and camera data that enables their system to dynamically adjust traffic lights and enhance traffic flow during consumption of less fuel.

Naramdeo et al. (2025) [6] developed a sensor-based intelligent traffic control that modulates signal timing based on the density of vehicles and identifies emergency vehicles. They performed better in terms of their approach than traditional fixed-time signal systems.

Laanaoui et al. (2024) [7] put emphasis on anomaly detection and load balancing in real time in traffic systems. Their machine learning-based model recognizes abnormal traffic patterns and offers optimum routing solutions to alleviate congestion.

A decentralized traffic prediction framework was suggested by Raghunath et al. (2024) [8] with the help of Temporal Convolutional Networks with Federated Learning. This model enables use of several traffic nodes to jointly process data without exchanging raw traffic data, a factor that guarantees privacy and enhances prediction quality.

The study by Fadila et al. (2024) [9], is an extensive review of smart technologies in traffic management, focusing on the combination of AI, IoT, blockchain, and intelligent transportation systems to improve mobility and sustainability in the city.

Wang et al. (2024) [10] proposed a model-based systems engineering approach to a cyber-physical system architecture to control traffic at urban crossroads. Its system incorporates sensing, communication, and computation to enhance traffic coordination and reliability.

Xu et al. (2023) [11] created a digital twin, smart mobility platform that is based on the cloud. Their platform integrates IoT sensors with advanced analytics to deliver traffic conditions real-time monitoring, prediction, and optimization.

In Jin et al. (2022) [12], the authors suggested an agent-based approach to traffic management that involves a multi-agent framework to dynamically manage traffic at the intersection. This will enhance the coordination and decision making between the traffic signals.

Oliveira et al. (2021) [13] developed a wireless communication network-based traffic monitoring system in real-time. Their system allows the centralized control of traffic lights and enhances the coordination and congestion.

Nagapragna and Joshi [14] designed a low cost traffic controller system based on Arduino and infrared sensors. Their embedded system can be used to adjust traffic lights depending on the number of vehicles, which offers scalability to a city.

The RFID-based traffic management system proposed by Lanke and Koul (2013) [15] dynamically adjusts the traffic lights by sensing the frequency of vehicles. Their strategy will enhance traffic efficiency and improve congestion detection.

All in all, the current literature has been of great advancement in predicting and controlling traffic. A majority of the systems however concentrate on either traffic management or parking solutions on their own. This is where there is a necessity of an integrated system that would integrate both capabilities, which is discussed in the proposed work.

III. OBJECTIVES

The main objective of the study is to design and develop a smart AI-based traffic and parking management system that enhances traffic flow and optimizes parking usage based on machine learning and computer vision. The system aims at making predictions of real time traffic congestion, making efficient decisions on traffic control and offering intelligent solutions on managing parking by comparing past and present traffic data.

Machine learning models like the Random Forest model are used to provide congestion prediction and ensure that the prediction is accurate by using the data provided by IoT sensors, traffic cameras, GPS sources, and historical traffic data. According to these forecasts, the system allows the adoption of dynamic traffic control measures and optimal route recommendations in order to minimize delays and congestion. The major aims of the proposed system are:

A. Traffic Congestion Prediction

To create a predictive model that can estimate the level of congestion in the city based on the data that is obtained via traffic cameras, GPS signals, IoT devices, and past traffic history.

B. Real-Time Data Integration

To facilitate constant incorporation of real-time traffic information, it is necessary to allow the system to update itself in real time and react properly to unexpected events like accidents, peak hour traffic, and road closures.

C. Smart Parking Management

To use a computer vision-based parking detector system to detect the available parking space in real-time and enable the user to find and book parking spaces conveniently.

D. Adaptive Traffic Control

To develop a smart traffic control system that would automatically modify signal timing and recommend alternative routes depending on the forecasted traffic congestion issues in order to enhance the efficiency of the traffic.

E. Emergency Vehicle Prioritization

To reduce response time and improve safety To allow emergency vehicles to move faster by giving priority routing and dynamic clearing of traffic lights.

Scalability to Smart City Integration. To create an infrastructure capable of being integrated into smart city infrastructure, which can support future expandability and real-life implementation, and is scalable and cost-effective.

IV. METHODOLOGY

The construction of the offered intelligent traffic and parking management system has a systematic methodology based on the combination of machine learning, computer vision, and real-time data processing methods. The system is intended to track traffic conditions, anticipate the state of congestion and handle the parking in an automated manner.

The initial phase will be data collection in which real-time and historical traffic data will be collected through various sources including traffic cameras, IoT sensors, GPS, and the available traffic databases. This information is very beneficial in terms of the density of vehicles, their speed, the occupation of roads, and the traffic dynamics. The data that is collected is sent to a central processing unit where it is further analyzed.

The second stage involves data preprocessing whereby the raw data is made ready to train the model. This involves the noise removal, missing values, irrelevant information and extraction of relevant features like traffic density, peak-hour trends, and congestion indicators. The machine learning model takes in the processed data.

The third phase is concerned with predicting traffic congestion with the help of the Random Forest algorithm. This model is comprised of a number of decision trees that are trained on various subsets of the data. The trees individually give a prediction of the congestion levels and the final prediction is a summation of the tree outputs. This group method enhances precision and lessens overfitting. The model is trained based on historical information and is constantly updated with real-time information to provide valid predictions.

At the same time a parking recognition system is applied using OpenCV and live video streams to detect the parking position. This module uses the video frames to detect the available and occupied parking spaces in real time. The information identified is continuously updated and presented to users on a web based interface allowing effective management of parking.

Dynamic traffic control is also included in the system, whereby the anticipated level of congestion is taken into account to modify traffic signal timings and propose alternative routes. This dynamic system can minimize the wait time in the crossroads and enhance the traffic flow. Also, an emergency vehicle priority module provides an emergency vehicle with a high priority in order to move faster through optimal route and clearance of signals.

Lastly, a monitoring and feedback system is incorporated to measure system performance based on measures like accuracy of prediction and response time. The model is retrained with continuous feedback to improve it over time. This approach makes the system scalable, adaptive and effective in real-time traffic and parking management, which is applicable in applications of smart cities.

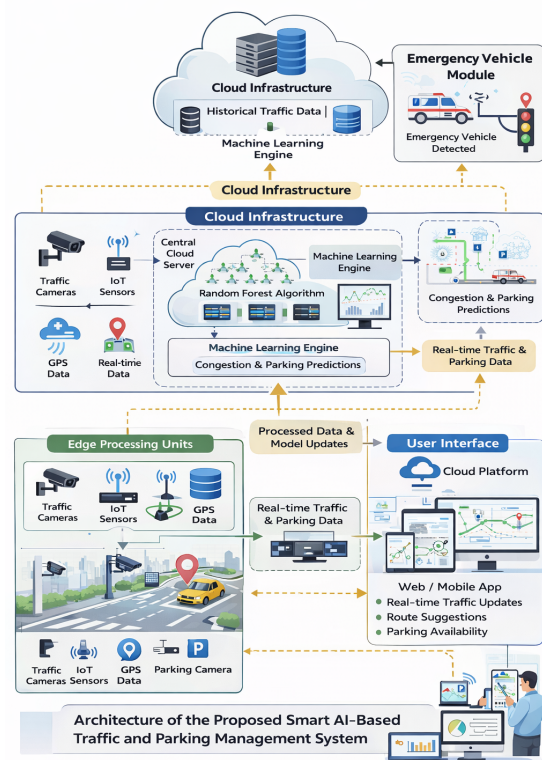


Fig. 1. Architecture of the Proposed Smart AI-Based Traffic and Parking Management System

At the same time, a parking management module which is built on computer vision will be present to monitor parking lots and identify vacant parking lots in real time to facilitate the smart use of space and convenient booking. The system also integrates dynamic traffic signal optimization and emergency vehicle priority to guarantee the implementation of a more efficient traffic flow and quicker emergency response. The system will be scalable and effective in managing transportation in modern smart cities due to its dynamically updated predictions and control strategies, done through continuous feedback and monitoring the performance. Moreover, the

Smart AI-Based Traffic and Parking Management System that is suggested to be implemented requires the incorporation of cloud-computing infrastructure and edge processing units. The edge devices, which are traffic cameras and IoT sensors that are mounted at road crossings and parking spaces, process real-time data and do preliminary preprocessing before sending the processed data to the central cloud server. This design minimizes latency and guarantees quicker response to traffic signal adjustments and parking detection. Cloud platform also saves huge amounts of historical traffic information and allows continuous model training and performance enhancement due to scalable computing resources.

Besides this, the system has a user interaction module that will enable commuters and traffic authorities to have real-time traffic and parking information by a mobile or web based application. The interface shows the level of congestion, parking spaces, optimal routes, and estimated time of travel. The system assists in eliminating needless vehicle traffic by giving information to the drivers beforehand, thereby enhancing the efficiency of the traffic in general.

Predictive analytics is a technology that can be integrated with intelligent traffic control to make proactive decisions in urban traffic management. The system also predicts traffic behavior and responds dynamically to changes in traffic lights and routes to accommodate congestion as it occurs, rather than responding to congestion as it happens. This predictive ability greatly minimizes the time spent in intersection delays and avoids the occurrence of serious congestion during peak hours.

In addition, the system is friendly to expansion and flexibility, and can be deployed both in large metropolitan settings and in small urban settings. The modular design enables the introduction of other modules like environmental monitoring sensors, integration of public transportation data and accident detection systems in the future developments. This adaptability is going to make the system adapt as the technologies of smart cities develop.

Altogether, the suggested methodology offers a full-fledged approach to the intelligent traffic and parking management, which integrates machine learning routines, computer vision strategies, and real-time data processing. The system improves traffic mobility, optimized parking use, and leads to the creation of sustainable and effective smart city transportation systems through efficient monitoring, predictive modeling and adaptive control.

V. RESULT ANALYSIS AND VALIDATION

A. Result Analysis

The effectiveness of the proposed intelligent traffic and parking management system is considered through the evaluation of the system in terms of its capability to correctly predict traffic congestion and enhance the traffic flow and parking utilization. The main task of the analysis is to evaluate the effectiveness of the system in estimating the level of congestion and optimizing the work of the urban traffic. Quantitative and qualitative assessments are both taken into account with major

aspects being the reduction of the traveling time, consumption of fuel, delays of the traffic, and the environment.

The system is evaluated based on a mix of the historical traffic databases and real-time data obtained via simulated IoT sensors, traffic cameras, and GPS-based sources. The findings suggest that the prediction model of congestion using the random forest is able to give consistent and reliable predictions of congestion in different traffic conditions.

B. Traffic Prediction Accuracy

Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and coefficient of determination (R^2) are standard measures of machine learning model performance. These indicators show how much congestion is predicted versus actual.

The Random Forest model has a high prediction accuracy because of its ensemble learning property, and a low probability of overfitting. Also, a confusion matrix is employed in order to assess the performance of classification under various traffic conditions, including low, moderate, and high congestion rates. The findings indicate good accuracy of classification and sound detection of congestion patterns.

C. Traffic Congestion Reduction

Before and after implementation, the efficiency of the system in decreasing congestion is measured by comparing the condition of traffic. Key performance indicators are average vehicle speed, queue length at the intersections and traffic density.

The findings indicate that the levels of congestions are reduced considerably, queues are shorter and waiting time is less, particularly during peak time. The system provides improved road network distribution by dynamically controlling the timing of the traffic signals and proposing alternative routes.

sub-topic: Travel Time and Route Optimization.

The time spent travelling is also enhanced by evaluating the average time spent in reaching the destinations in both congested and optimized conditions. The system uses real-time forecasts to suggest effective paths and re-time the signals to align with the paths.

Through the experiment, there has been a visible difference in the travel time, which enhances the convenience of commuters and the overall transport efficiency. The less time wastage at the intersection also helps in the improvement of traffic flow and the driving experience.

D. Parking Management Performance

The parking detection system is determined by the computer vision on its capability to detect empty and occupied parking spaces. The system is effective in varying lighting conditions and environmental conditions and is highly accurate in detection.

The information on the availability of parking spots in real-time helps users to find parking spots easily, decrease unnecessary traffic and reduce congestion in parking lots.

E. Environmental Impact

The system environmental advantages are evaluated by evaluating the decrease in fuel consumption and carbon emissions. The enhanced movement and wait time of the traffic in the intersections will result in less fuel consumption and the decrease in emissions.

These findings demonstrate the opportunities of smart traffic and parking systems to facilitate sustainable urban transport and help to make the environment greener.

F. Validation

In order to confirm the workability of the suggested system, simulation testing and pilot level testing are performed. The system performance is compared to the traditional fixed-time traffic signal systems.

The performance is measured using evaluation metrics, including accuracy, F1-score, confusion matrix, and error rate. The findings confirm the effectiveness of the suggested system in terms of its ability to predict more accurately and handle congestion, which proves its reliability and feasibility in practice.

TABLE I
SUMMARY OF RESULT ANALYSIS

Evaluation Metric	Description	Expected Outcome
Traffic Congestion Prediction Accuracy	Accuracy of ML model (Random Forest / AI model) in predicting congestion levels using historical and real-time traffic data	High prediction accuracy with low MAE, RMSE, and high R^2 score
Parking Slot Detection Accuracy	Accuracy of computer vision model in detecting available parking spaces from camera feeds	Precise detection with minimal false positives and false negatives
Traffic Signal Optimization	Evaluation of adaptive signal timing based on congestion levels	Reduced waiting time and smoother vehicle flow at intersections
Average Travel Time Reduction	Comparison of travel time before and after system implementation	Noticeable reduction in travel time during peak hours
Queue Length Reduction	Measurement of vehicle queue length at major junctions	Decrease in average queue length after AI-based control
Emergency Vehicle Prioritization	Dynamic adjustment of signals/routes for emergency vehicles	Faster clearance and reduced response time
Parking Utilization Efficiency	Optimized parking allocation and reservation system analysis	Better parking space usage and reduced search time
Environmental Impact	Measurement of fuel consumption and emission levels	Reduced fuel usage and lower carbon emissions

The bar graph evaluates the key performance indicators of the traffic performance in the pre-implementation period and in the post-implementation period of the proposed AI-based traffic management system. It has been found out that the average vehicle speed has increased significantly, as well as the queue length and the traffic density has been decreasing. This implies that the system has been successful to

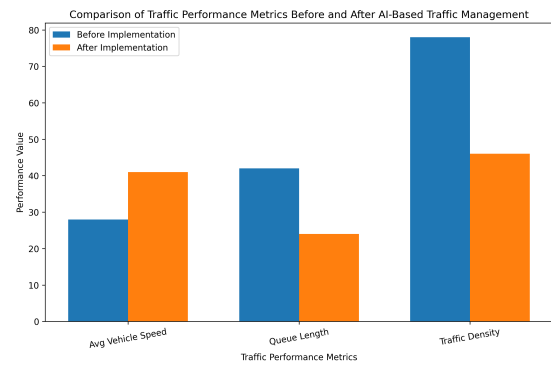


Fig. 2. Comparison of traffic performance metrics before and after AI-based traffic management implementation.

maximize traffic flow, reduce congestion and general efficiency of transportation. The line graph shows how the emission

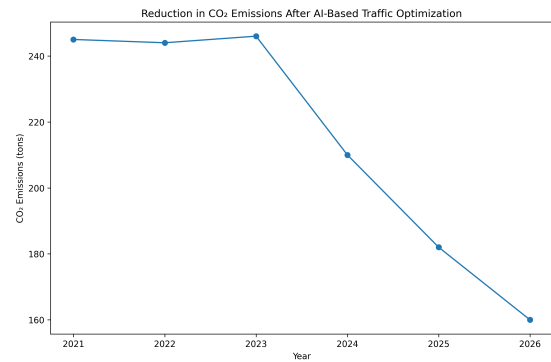


Fig. 3. Reduction in CO emissions over time after implementing the AI-based traffic optimization system.

of CO₂ decreased over the years after the implementation of the AI-based traffic optimization system. An overall reduction can be seen after the implementation of a system although the amount of emission does not change prior to the implementation. This goes to show that increased traffic movement and decreased congestion lead to the lessening of fuel usage and environmental effect. The trend presented in the graph brings out the environmental sustainability aspect of AI-based traffic management systems. Other than congestion minimization, the suggested model can also benefit the smart cities efforts, advancing cleaner transportation, enhancing air quality, and decreasing the total carbon imprint of the mobility systems in the cities. The confusion matrix shows classification effectiveness of the proposed traffic congestion prediction model in three classes Free Flow, Moderate and Congested. Large values in the diagonal mean correct prediction and small off-diagonal values mean that the misclassification is minimal. The obtained results demonstrate that the model is rather successful in prediction accuracy and reliability in real-time classification of the state of traffic.

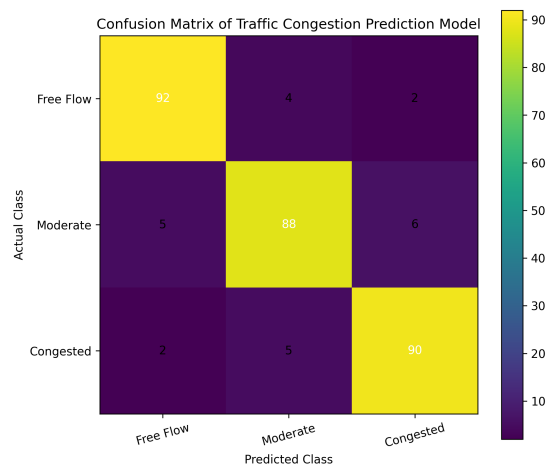


Fig. 4. Confusion matrix representing the performance of the proposed traffic congestion prediction model.

VI. CONCLUSION

This paper introduces a smart traffic and parking management system that combines machine learning and computer vision to overcome the issues of congestion and ineffective parking in urban areas. The proposed system is based on real-time and historical traffic data that enable to accurately predict the level of congestion and optimize the work of traffic signals through the Random Forest algorithm. Moreover, the computer vision-based parking module offers real-time data on parking spots, which can assist users to find parking spots fast and minimize unwanted vehicle movements.

The experimental findings show that the system greatly enhances the accuracy of traffic prediction, minimizes traveling time, minimizes fuel consumption, and decreases environmental impact. Prioritization of the emergency vehicles also adds value to the system, as it allows responding more quickly and ensuring the safety of the roads in case of emergency.

The proposed system is more flexible, scalable and real-time responsive as compared to traditional traffic management techniques that are based on fixed or less adaptive mechanisms. In general, the system offers an intelligent, cost-efficient, and scalable system to modern transportation in cities. It helps in establishing effective, sustainable, and smart mobility systems by integrating traffic forecasting, parking, and automation in a comprehensive platform.

VII. FUTURE WORK

The intelligent traffic and parking management system that is suggested can be improved further by adding more sophisticated and extended technologies to accommodate the next-generation smart transportation systems. The combination of deep learning and reinforcement learning approaches is one of the promising directions as it can enhance the accuracy of the predictions and provide more adaptive traffic light control. Particularly, reinforcement learning can aid the system in its continuous learning based on the current traffic conditions and decision-making process optimization as time goes on.

Moreover, further enhancements may involve the adoption of more sophisticated IoT-related sensors, connected cars and mobile GPS data to gather a greater variety of more valuable traffic data. The system can further be integrated to accommodate vehicle-to-infrastructure (V2I) communication to enable more coordination between traffic signals and smart vehicles and hence enhancing safety and improving on congestion.

Additionally, the system can be scaled to a bigger capacity by creating real-time mobile applications that will give users real-time traffic details, the best route options, and where to park. Enhancements in parking automation, such as automated parking guidance and integration with electric vehicle charging stations, can also be considered.

Future studies could aim at enhancing the precision of prediction based on big data of real world and integration of multi-source data. These developments will enhance the capacity of the system to facilitate smart city programs, encourage sustainable transportation and facilitate smarter, more efficient, and environmentally friendly urban mobility solutions.

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